

Lab 5 - Introduction to Flex

Q1.

```
%{
```

```
int v_count = 0;
```

```
int c_count = 0;
```

```
%}
```

```
%%
```

```
[aeiouAEIOU] { v_count++; }
```

```
[a-zA-Z] { c_count++; }
```

```
.\n { /* Skipping char */ }
```

```
%%
```

```
int main() {
```

```
    printf("Enter text: ");
```

```
    yylex();
```

```
    printf("\nVowels: %d\nConsonants: %d\n", v_count, c_count);
```

```
    return 0;
```

```
}
```

```
int yywrap() { return 1; }
```

Output:

```
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ flex q1.1
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ gcc lex.yy.c -o abc
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ ./abc
Enter text: hi
hello
bye
why

Vowels: 4
Consonants: 9
```

Q2.

```
%{
```

```
    int chr_total = 0, wrd_total = 0, gap_total = 0, row_total = 0;
```

```
%}
```

```
%%
```

```
\n    { row_total++; chr_total++; }
```

```
" "    { gap_total++; chr_total++; }
```

```
[^\\t\\n ]+ { wrd_total++; chr_total += yyleng; }
```

```
.      { chr_total++; }
```

```
%%
```

```
int main() {
```

```
    yylex();
```

```
    printf("\\nRows: %d\\nWords: %d\\nBlanks: %d\\nTotal Chars: %d\\n",
row_total, wrd_total, gap_total, chr_total);
```

```
    return 0;
```

```
}
```

```
int yywrap() { return 1; }
```

Output:

```
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ flex q2.1
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ gcc lex.yy.c -o abc
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ ./abc
hi
hello
bye
why

Rows: 4
Words: 4
Blanks: 0
Total Chars: 17
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ █
```

Q3.

```
%{
```

```
    int pos_int = 0, neg_int = 0, posflt = 0, negflt = 0;
```

```
%}
```

```
%%
```

```
[-][0-9]+      { neg_int++; }
```

```
[+]?[0-9]+     { pos_int++; }
```

```
[-][0-9]*\.[0-9]+ { negflt++; }
```

```
[+]?[0-9]*\.[0-9]+ { posflt++; }
```

```
.\n           { }
```

```
%%
```

```

int main() {
    yylex();

    printf("\nPos Int: %d\nNeg Int: %d\nPos Float: %d\nNeg Float: %d\n",
pos_int, neg_int, posflt, negflt);

    return 0;
}

```

```

int yywrap() { return 1; }

```

Output:

```

6CSEC1@db1-23:~/Documents/CD_230905152/L5$ flex q3.1
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ gcc lex.yy.c -o abc
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ ./abc

```

```

12
-12
12.1
-12.1

```

```

Pos Int: 1
Neg Int: 1
Pos Float: 1
Neg Float: 1

```

Q4.

```

%{
    int sc_hits = 0;
    int pr_hits = 0;
}%

%%

"scanf" { sc_hits++; printf("READ"); }

```

```
"printf" { pr_hits++; printf("WRITE"); }
```

```
.\n    { printf("%s", yytext); }
```

```
%%
```

```
int main() {
```

```
    yylex();
```

```
    fprintf(stderr, "\nScanned: %d | Printed: %d\n", sc_hits, pr_hits);
```

```
    return 0;
```

```
}
```

```
int yywrap() { return 1; }
```

Output:

```
6CSEC1@dbl-23:~/Documents/CD_230905152/L5$ flex q4.1
6CSEC1@dbl-23:~/Documents/CD_230905152/L5$ gcc lex.yy.c -o abc
6CSEC1@dbl-23:~/Documents/CD_230905152/L5$ ./abc
```

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int a = 10; int b = 20;
    int s;
    scanf("%d",s);
    printf("Hello World");
    return 0;
}
#include <stdio.h>
#include <stdlib.h>
```

```
int main() {
    int a = 10; int b = 20;
    int s;
    READ("%d",s);
    WRITE("Hello World");
    return 0;
}
```

Scanned: 1 | Printed: 1

Q5.

```
%{ /* Decimal to Hex */ %}
```

```
%%
```

```
[0-9]+ { int val = atoi(yytext); printf("%X", val); }
```

```
.\n { printf("%s", yytext); }
```

```
%%
```

```
int main() {
    yylex();
```

```

        return 0;

}

int yywrap() { return 1; }

```

Output:

```

6CSEC1@db1-23:~/Documents/CD_230905152/L5$ flex q5.1
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ gcc lex.yy.c -o abc
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ ./abc
16
10
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ █

```

Q6.

```

%{

#include <ctype.h>

%}

%%

"/".* { printf("%s", yytext); }

"/"([^\*]|\\*+[^/])*(\*+/" ) { printf("%s", yytext); }

[A-Z] { printf("%c", tolower(yytext[0])); }

.\n { printf("%s", yytext); }

%%

int main() {

    yylex();

```

```
        return 0;

}

int yywrap() { return 1; }
```

Output:

```
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ flex q6.1
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ gcc lex.yy.c -o abc
6CSEC1@db1-23:~/Documents/CD_230905152/L5$ ./abc
//This is comment
//This is comment
HELLOhello
hellohello
```